

PixCuboid: Room Layout Estimation from Multi-view Featuremetric Alignment

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Summary

- We propose a novel multi-view room layout estimation method based on featuremetric alignment.
 - Input:** A collection of posed images.
 - Output:** The predicted room layout (a cuboid).
- Using multiple views resolves the global scale.
- With our flexible, optimization-based approach we can:
 - Easily change the cost function.
 - Do multi-room layout estimation.
 - Refine other geometries, not just cuboids.

Datasets

We create two datasets, based on ScanNet++ v2 and 2D-3D-Semantics, to benchmark multi-view cuboid room layout estimation.

- Ground truth layouts:** For each scene, a cuboid is fitted to vertices labeled “floor”, “wall” and “ceiling” in the semantic mesh. We manually inspect the results and discard non-cuboid scenes.
- Images:** For ScanNet++ we sample image tuples, consisting of ten random DSLR images, to use as input. In 2D-3D-Semantics we randomly pick two panoramas, each split into four perspective images, for every room.

Dataset	Train	Validation	Test
ScanNet++ v2	391	10	18
2D-3D-Semantics	0	0	160

Table 1. Number of scenes for the two datasets.

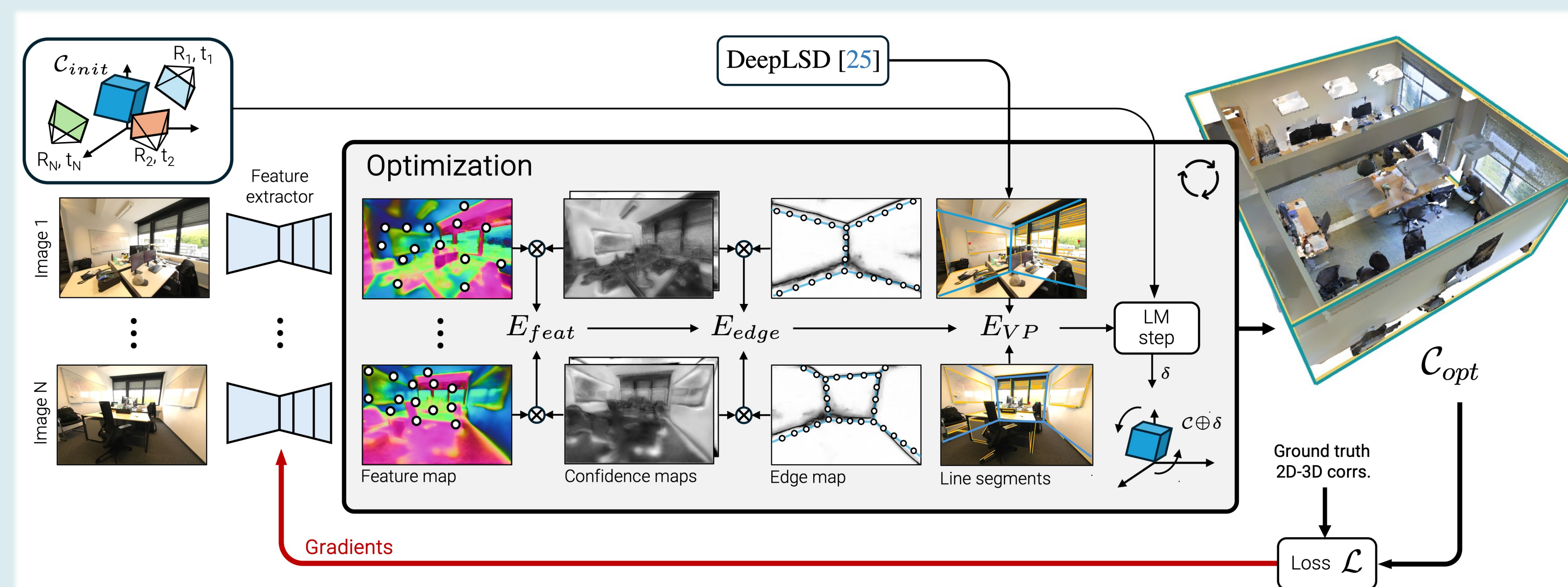


Figure 1. Taking only posed images as input, PixCuboid estimates the room layout $\mathcal{C}_{opt}$ by optimizing from a coarse initial estimate $\mathcal{C}_{init}$. First, a deep network predicts dense feature-, confidence- and edge maps for each image. The optimization then minimizes a combination of three terms (see below). The network is trained end-to-end by supervising on the result of the optimization, propagating gradients back to the network through the optimization steps.

Cuboid Optimization

To refine the cuboid $\mathcal{C}$ we define a cost function $E(\mathcal{C})$ depending on three terms:

- A featuremetric cost $E_{feat}(\mathcal{C}) = \sum_{i,j} \sum_k w_{ijk} \rho(\| \mathbf{F}_i[\mathbf{x}_{ik}] - \mathbf{F}_j[\mathcal{W}_{i \rightarrow j}(\mathbf{x}_{ik}, \mathcal{C})] \|^2)$.
- A monocular edge cost $E_{edge}(\mathcal{C}) = \sum_i \sum_j w_{ij} \mathbf{E}_i[\Pi_i(\mathbf{R}_i \mathbf{X}_j + \mathbf{t}_i)]^2$.
- A VP-based cost $E_{VP}(\mathcal{C}) = \sum_i \sum_j \min \left(\min_{k \in \{1,2,3\}} D_{VP}(\mathbf{l}_{ij}, \mathbf{v}_{ik}), \tau \right)^2$.

We minimize $E(\mathcal{C}) = E_{feat}(\mathcal{C}) + \alpha E_{edge}(\mathcal{C}) + \beta E_{VP}(\mathcal{C})$ using standard Levenberg-Marquardt optimization.

- Multi-scale:** The optimization is performed in a coarse-to-fine manner, with increasingly higher resolution feature maps $\mathbf{F}_i$ and edge maps $\mathbf{E}_i$.
- Initialization:** We use a simple heuristic to initialize $\mathcal{C}$ from the known camera poses.

Training

PixCuboid is trained on our ScanNet++ v2 training set with the loss

$$\mathcal{L}(\mathcal{C}) = \frac{1}{N} \sum_{i,j} \sum_k \|\mathcal{W}_{i \rightarrow j}(\mathbf{x}_{ik}^{GT}, \mathcal{C}) - \Pi_j(\mathbf{R}_j \mathbf{X}_{ik}^{GT} + \mathbf{t}_j)\|^2.$$

- $\{(\mathbf{x}_{ik}^{GT}, \mathbf{x}_{ik}^{GT})\}$: 2D-3D correspondences on the floor/wall/ceiling.

Results

We evaluate on 2D-3D-Semantics, comparing against both single- and multi-view methods (none of which are trained on this dataset).

Method	Input	IoU $\uparrow$	Chamfer $\downarrow$	Rotation $\downarrow$	Depth $\downarrow$	Normal $\uparrow$
Total3D	1 x 	31.8	1.57 m	8.9°	1.41 m	44.1
Implicit3D	1 x 	31.6	1.59 m	10.4°	1.51 m	37.6
Deep3DLayout	1 x 	58.8	0.68 m	N/A	0.44 m	52.6
LED ² -Net	1 x 	68.7	0.45 m	N/A	0.40 m	34.6
PSMNet	2 x 	43.5	1.04 m	N/A	0.63 m	51.4
PixCuboid (ours)	8 x 	89.0	0.18 m	0.7°	0.10 m	96.1

Table 2. Room layout estimation results on the cuboid-shaped spaces of 2D-3D-S.

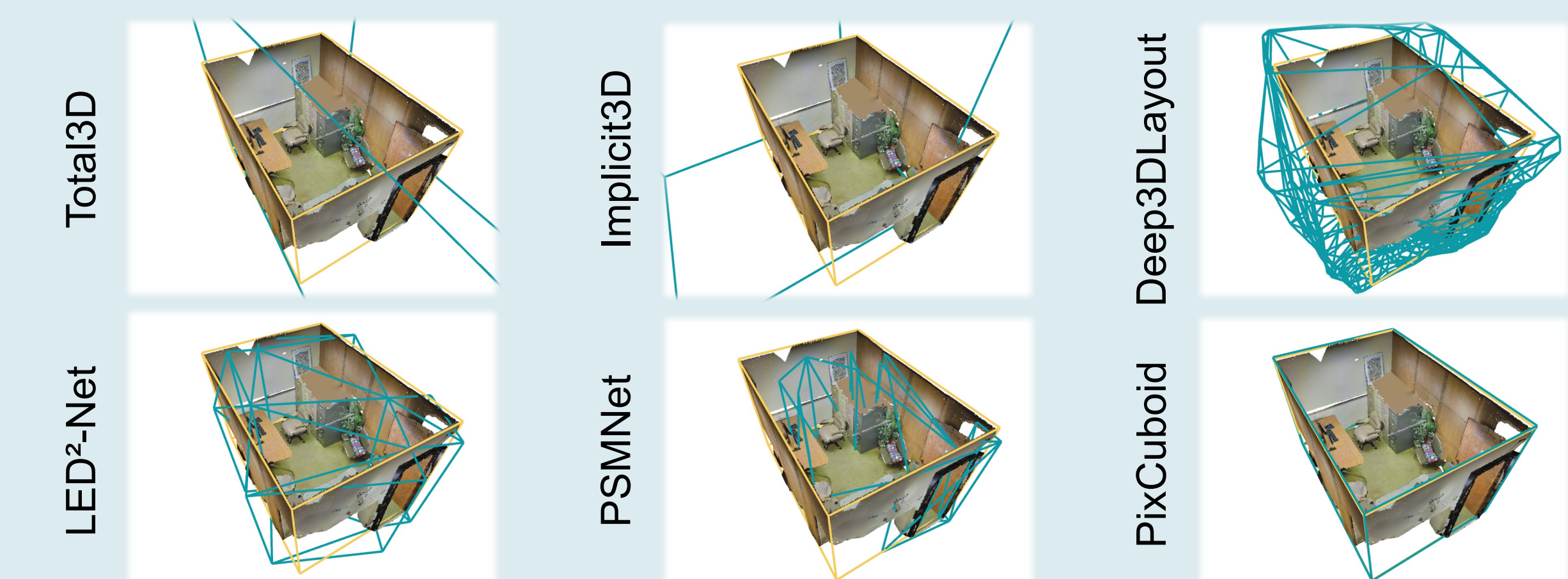


Figure 2. Qualitative comparisons of predicted room layouts for one space in 2D-3D-Semantics. Predictions are shown in blue and the ground truth cuboids in yellow.